

Kubernetes ConfigMap Not Updating SOP

Objective

Ensure application pods consume the latest ConfigMap values when changes are not reflected after a configuration update.

Scope

Applicable to Kubernetes workloads, platform operations, DevOps teams, SRE teams, application teams, and synthetic incident workflows across development, staging, and production-like environments.

Pre-requisites

1. Access to Kubernetes cluster using kubectl and the correct namespace context.
2. Access to application, pod, deployment, service, ingress, and node logs where applicable.
3. Access to monitoring dashboards such as Grafana, Prometheus, cloud monitoring, or container insights.
4. Permission to inspect manifests, secrets, configmaps, RBAC bindings, PVCs, and rollout history.
5. Escalation contact for DevOps, platform, infrastructure, network, and application teams.

Incident Metadata

Field	Value
Ticket ID	T021
Issue	Kubernetes configmap not updating
Category	Configuration
Service	Kubernetes
Severity	Low
Status	Resolved
Description	Changes not reflected in pods
Expected Resolution	Restart pods after config update

Procedure

1. Confirm the ConfigMap name, namespace, changed key, and expected value.
2. Check whether the pod consumes the ConfigMap as environment variables or as a mounted volume.
3. Inspect the live ConfigMap data using `kubectl get configmap`.
4. Validate pod spec references the correct ConfigMap and key names.
5. Restart pods if ConfigMap values are injected as environment variables.
6. For volume-mounted ConfigMaps, wait for kubelet sync and verify mounted files.
7. Roll out the deployment restart when required.
8. Confirm application logs reflect the new config and update ticket evidence.

Common Commands

```
kubectl config current-context
kubectl get pods -n <namespace> -o wide
kubectl describe pod <pod-name> -n <namespace>
kubectl logs <pod-name> -n <namespace> --tail=100
kubectl get events -n <namespace> --sort-by=.lastTimestamp
kubectl get deploy,svc,ingress,pvc -n <namespace>
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Config not reflected	Pod env vars not refreshed	Restart pods or rollout deployment
Wrong value used	Incorrect ConfigMap key reference	Fix key name in manifest
Mounted file stale	Kubelet sync delay	Wait or restart pod if required
ConfigMap not found	Wrong namespace or name	Correct reference and namespace

Best Practices

1. Version critical configuration changes.
2. Restart workloads after env-var ConfigMap updates.
3. Avoid manual production config edits where possible.
4. Document config ownership and rollout process.

References

[Kubernetes Docs](#) | [Internal Tickets CSV](#) | [Cluster Events](#) | [Monitoring Dashboards](#) | [Application Logs](#) | [Platform Runbook](#)